

# Solubility Equilibria

Answer key with worked solutions

## A. Expressions

### 1. K<sub>sp</sub> expressions.

- (a)  $\text{BaSO}_4(\text{s}) \rightleftharpoons \text{Ba}^{2+} + \text{SO}_4^{2-}; K_{\text{sp}} = [\text{Ba}^{2+}][\text{SO}_4^{2-}]$   
 (b)  $\text{Mg}(\text{OH})_2(\text{s}) \rightleftharpoons \text{Mg}^{2+} + 2 \text{OH}^{-}; K_{\text{sp}} = [\text{Mg}^{2+}][\text{OH}^{-}]^2$   
 (c)  $\text{Ag}_2\text{CrO}_4(\text{s}) \rightleftharpoons 2 \text{Ag}^{+} + \text{CrO}_4^{2-}; K_{\text{sp}} = [\text{Ag}^{+}]^2[\text{CrO}_4^{2-}]$   
 (d)  $\text{Ca}_3(\text{PO}_4)_2(\text{s}) \rightleftharpoons 3 \text{Ca}^{2+} + 2 \text{PO}_4^{3-}; K_{\text{sp}} = [\text{Ca}^{2+}]^3[\text{PO}_4^{3-}]^2$

## B. Molar solubility

### 2. BaSO<sub>4</sub>.

$$s = \sqrt{1.1 \times 10^{-10}} = 1.0 \times 10^{-5} \text{ mol/L}; 1.0 \times 10^{-5} \text{ mol/L} \times 233.39 \text{ g/mol} = 2.4 \times 10^{-3} \text{ g/L}$$

### 3. Mg(OH)<sub>2</sub>.

- (a)  $K_{\text{sp}} = 4s^3$ , so  $s = \sqrt[3]{\frac{1.8 \times 10^{-11}}{4}} = 1.7 \times 10^{-4} \text{ mol/L}$   
 (b)  $[\text{OH}^{-}] = 2s = 2 \times 1.65 \times 10^{-4} = 3.3 \times 10^{-4} \text{ mol/L}$   
 (c)  $\text{pOH} = -\log(3.30 \times 10^{-4}) = 3.48$ ;  $\text{pH} = 14.00 - 3.48 = 10.52$

### 4. Ag<sub>2</sub>CrO<sub>4</sub>.

$$[\text{Ag}^{+}] = 2s, [\text{CrO}_4^{2-}] = s; K_{\text{sp}} = (2s)^2s = 4s^3 = 4 \times (6.5 \times 10^{-5})^3 = 1.1 \times 10^{-12}$$

### 5. Comparing.

AgCl:  $s = \sqrt{1.8 \times 10^{-10}} = 1.3 \times 10^{-5} \text{ mol/L}$ . Ag<sub>2</sub>CrO<sub>4</sub>:  $s = 6.5 \times 10^{-5} \text{ mol/L}$ , **five times larger**. Their K<sub>sp</sub> formulas differ ( $s^2$  versus  $4s^3$ ), so K<sub>sp</sub> values can only be compared directly for salts with the same ion ratio.

## C. Common ion and precipitation

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6.  $\text{CaF}_2$  in  $\text{NaF}$ .

$[\text{F}^-] \approx 0.010 \text{ mol/L}$ ;  $K_{\text{sp}} = s(0.010)^2$ , so  $s = \frac{3.9 \times 10^{-11}}{1.0 \times 10^{-4}} = 3.9 \times 10^{-7} \text{ mol/L}$ , about **550 times less** than in pure water.

7. Mixing.

Total volume 50.0 mL.

$$[\text{Ba}^{2+}] = 0.010 \text{ mol/L} \times \frac{10.0 \text{ mL}}{50.0 \text{ mL}} = 2.0 \times 10^{-3} \text{ mol/L}$$

$$[\text{SO}_4^{2-}] = 0.0050 \text{ mol/L} \times \frac{40.0 \text{ mL}}{50.0 \text{ mL}} = 4.0 \times 10^{-3} \text{ mol/L}$$

$Q = (2.0 \times 10^{-3})(4.0 \times 10^{-3}) = 8.0 \times 10^{-6}$ , far greater than  $K_{\text{sp}} = 1.1 \times 10^{-10}$ :  **$\text{BaSO}_4$  precipitates.**

## D. Challenge

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8. Milk of magnesia.

$\text{Mg}(\text{OH})_2(\text{s}) \rightleftharpoons \text{Mg}^{2+} + 2 \text{OH}^-$ . Stomach acid supplies  $\text{H}^+$ , which removes  $\text{OH}^-$  by forming water ( $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$ ). By Le Chatelier's principle, the equilibrium shifts to the **right** to replace the  $\text{OH}^-$ , so more  $\text{Mg}(\text{OH})_2$  dissolves, neutralizing the acid.